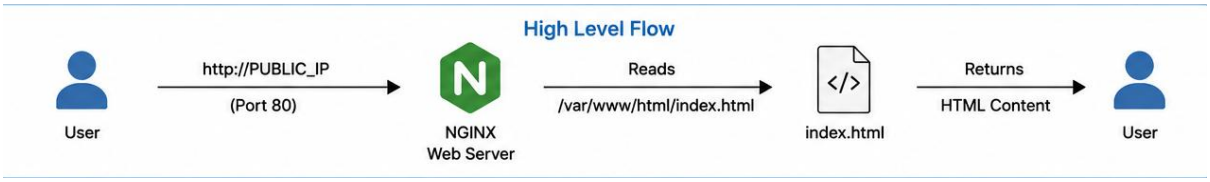
NGINX-WEBSERVER

NGINX is a high-performance web server that can be used for:

1. Static web hosting
2. Reverse proxy
3. Load balancing
4. SSL termination
5. Virtual hosting — one NGINX server can host multiple websites

---

HOSTING STATIC WEBSITE ON NGINX



1. Install NGINX

1. sudo apt update
2. sudo apt install nginx -y

2. Start & Enable NGINX

1. sudo systemctl start nginx
2. sudo systemctl enable nginx

4. Check NGINX status

1. sudo systemctl status nginx

5. Go to NGINX web directory & create index.html file

1. cd /var/www/html/
2. sudo vim index.html
   <h1> Hello NGINX! </h1>

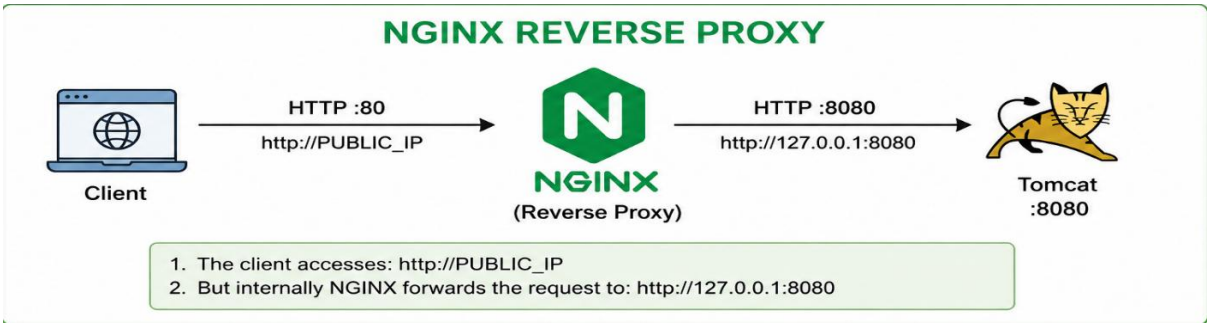
7. Restart NGINX

1. sudo systemctl restart nginx

8. Test the website

1. http://PUBLIC_IP

REVERSE PROXY ON NGINX



Install Java

1. sudo apt update
2. sudo apt install openjdk-17-jdk -y
3. java -version

Download & Start Tomcat

1. wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.120/bin/apache-tomcat-9.0.120.tar.gz
2. tar -xzf apache-tomcat-9.0.120.tar.gz
3. cd apache-tomcat-9.0.120/bin/
4. ./startup.sh
5. Check Tomcat:
   http://PUBLIC_IP:8080

So suppose Tomcat is running on: http://PUBLIC_IP:8080

Configure NGINX Reverse Proxy

1. vim /etc/nginx/sites-available/reverse-proxy

```
server {
    listen 80;
    server_name _;

    location / {
        proxy_pass http://localhost:8080;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

Enable the Configuration

2. sudo ln -s /etc/nginx/sites-available/reverse-proxy /etc/nginx/sites-enabled/
3. Remove the default configuration if required:
   sudo rm /etc/nginx/sites-enabled/default

Test NGINX Configuration

1. sudo nginx -t

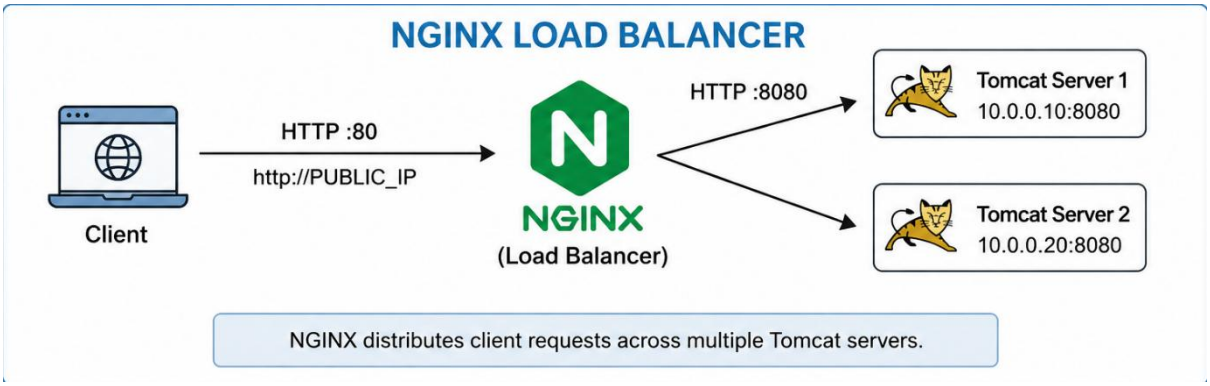
Restart NGINX

2. sudo systemctl restart nginx

Test Reverse Proxy

1. http://PUBLIC_IP
   NGINX will receive the request on port 80 and forward it to: http://127.0.0.1:8080

LOAD BALANCING ON NGINX



1. Tomcat Server 1 (http://10.0.0.10:8080)
2. Tomcat Server 2 (http://10.0.0.20:8080)
3. NGINX Load Balancer (http://PUBLIC_IP)

Configure NGINX Load Balancer

1. vim /etc/nginx/sites-available/load-balancer

```
upstream backend {
   server 10.0.0.10:8080;
   server 10.0.0.20:8080;
}

server {
   listen 80;
   server_name _;

   location / {
      proxy_pass http://backend;
      include /etc/nginx/proxy_params;
   }
}
```

Enable Configuration & Delete a default file

1. sudo ln -s /etc/nginx/sites-available/load-balancer /etc/nginx/sites-enabled/

2. sudo rm /etc/nginx/sites-enabled/default

Test Configuration

1. sudo nginx -t

Restart NGINX

1. sudo systemctl restart nginx

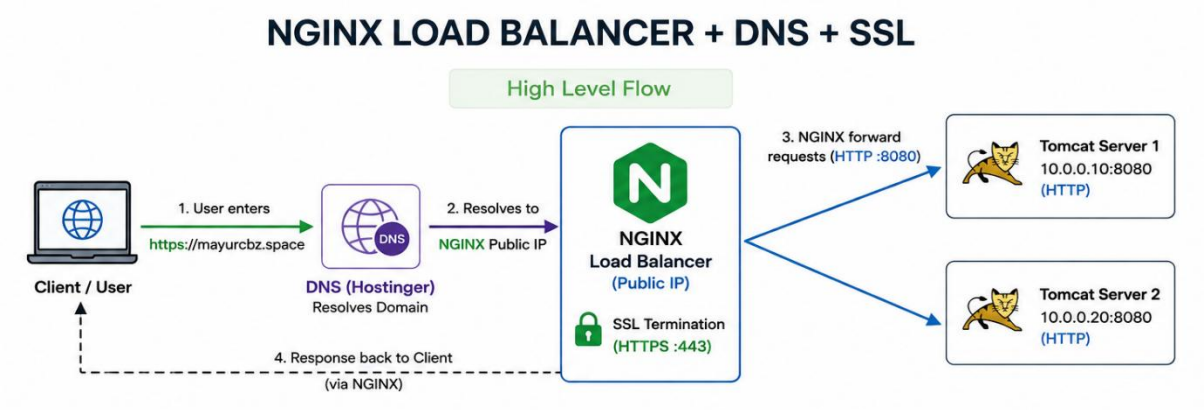
Test Load Balancing

1. http://PUBLIC_IP

Note:

1. NGINX distributes requests between:
   Tomcat Server 1
   Tomcat Server 2
2. By default, NGINX uses a round-robin style distribution for an upstream with multiple servers.

LOAD BALANCER + DNS + SSL



Configure NGINX Load Balancer

1. vim /etc/nginx/sites-available/load-balancer

```
upstream backend {
   server 10.0.0.10:8080;
   server 10.0.0.20:8080;
}

server {
   listen 80;
   server_name mayurcbz.space www.mayurcbz.space;

   location / {
      proxy_pass http://backend;
      include /etc/nginx/proxy_params;
```

```
    }
}
```

Enable Configuration & Delete default file

1. ln -s /etc/nginx/sites-available/load-balancer /etc/nginx/sites-enabled/
2. rm -rf /etc/nginx/sites-enabled/default

Test NGINX

1. sudo nginx -t

Restart NGINX

1. sudo systemctl restart nginx

Test NGINX Load Balancer

1. http://PUBLIC_IP

Configure DNS on Hostinger:

Add an A record:

1. Type: A
2. Name: @
3. Value: NGINX-PUBLIC-IP
4. TTL: 300

For www:

1. Type: A
2. Name: www
3. Value: NGINX-PUBLIC-IP
4. TTL: 300

Test Domain

1. http://mayurcbz.space

CONFIGURE SSL CERTIFICATE ON NGINX

We can use Let's Encrypt to get a free SSL certificate.

Install Certbot

1. sudo apt update
2. sudo apt install certbot python3-certbot-nginx -y

Generate SSL Certificate

1. sudo certbot --nginx
   Enter your email:
   Accept the terms.
   Redirect HTTP to HTTPS

Check Certificates

1. certbot certificates

Restart NGINX

1. sudo systemctl restart nginx

Check Port 80 & 443

1. lsof -i :80
2. lsof -i :443

Test HTTPS

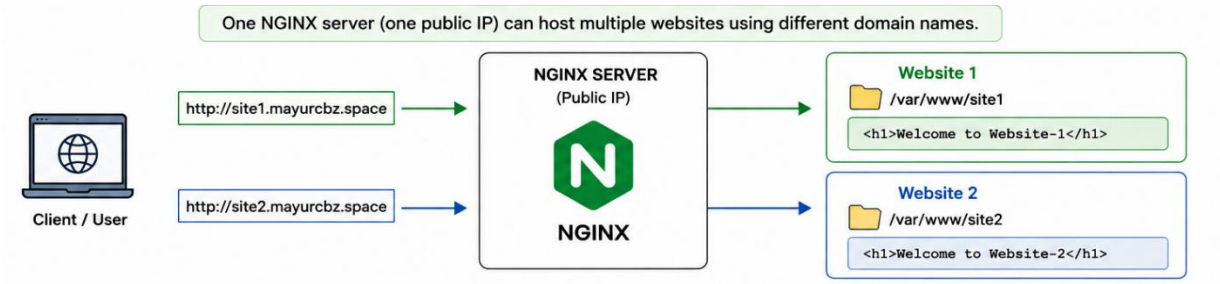
1. https://mayurcbz.space

CERTIFICATE RENEWAL

1. sudo certbot renew --dry-run
   If successful, Certbot's automatic renewal mechanism can renew the certificate when necessary.

VIRTUAL HOSTING ON NGINX

One NGINX server can host multiple websites using different domain names.



Install NGINX

1. sudo apt update
2. sudo apt install nginx -y

Start & Enable NGINX

1. sudo systemctl start nginx
2. sudo systemctl enable nginx

Check NGINX

1. curl localhost

CREATE WEBSITE 1

1. mkdir -p /var/www/site1
2. vim /var/www/site1/index.html
   <h1>Welcome to Website-1</h1>

CREATE WEBSITE 2

1. mkdir -p /var/www/site2
2. vim /var/www/site2/index.html
   <h1>Welcome to Website-2</h1>

CREATE VIRTUAL HOST FOR WEBSITE 1

1. vim /etc/nginx/sites-available/site1

```
server {
   listen 80;

   server_name site1.mayurcbz.space;
   root /var/www/site1;
   index index.html;

   location / {
      try_files $uri $uri/ =404;
   }
}
```

CREATE VIRTUAL HOST FOR WEBSITE 2

1. vim /etc/nginx/sites-available/site2

```
server {
   listen 80;

   server_name site2.mayurcbz.space;

   root /var/www/site2;
   index index.html;

   location / {
      try_files $uri $uri/ =404;
   }
}
```

ENABLE WEBSITE 1

1. ln -s /etc/nginx/sites-available/site1 /etc/nginx/sites-enabled/

ENABLE WEBSITE 2

1. ln -s /etc/nginx/sites-available/site2 /etc/nginx/sites-enabled/

TEST NGINX CONFIGURATION

1. sudo nginx -t

RESTART NGINX

1. sudo systemctl restart nginx

DNS CONFIGURATION ON HOSTINGER

WEBSITE 1

Add an A record:

1. Type: A
2. Name: site1
3. Value: NGINX-PUBLIC-IP
4. TTL: 300

WEBSITE 2

Add an A record:

1. Type: A
2. Name: site2
3. Value: NGINX-PUBLIC-IP
4. TTL: 300

TEST THE WEBSITES

Website 1

1. http://site1.mayurcbz.space

Website 2

1. http://site2.mayurcbz.space
